Inorganic Cume September 15, 2000

Metal carbonyl chemistry

1) Give an example of each of the following classes of metal – CO species. Your example must be consistent with the electron counting rules for organometallic complexes.

   a) neutral mononuclear complex with [CO/metal] not equal to five.
   b) heterometallic binuclear metal carbonyl
   c) carbonyl hydride
   d) carbonylate anion
   e) cyclopentadienyl carbonyl

2) Give a reasonable synthetic scheme for each of the following:

   a) $Mn(CO)_5Br$
   b) $(OC)_4Co - Re(CO)_5$
   c) $Cr(CO)_5PEt_3$ , without formation of *bis* - phosphine impurities
   d) $Fe_2(CO)_9$ and $Fe_3(CO)_{12}$

3) Predict the initial products for the following reactions:

   a) $H_2O + Fe(CO)_5 \longrightarrow$

   b) $Me_3N{\rightarrow}O$ + $\longrightarrow$

4) One type of CO to metal bonding pattern is referred as a terminal carbonyl. Draw sketch that indicates the nature of a terminal CO ligand. What other common types of M –CO bonding are commonly observed and how are they distinguished. (*Be specific as to technique you would use and expected experimental observations.*)

5) The binuclear species $Cr_2(CO)_8$ exhibits two isomeric forms. Draw these two structures and suggest how you could easily distinguish between these two forms.